Unpleasant Monetarist Arithmetic

Discrete time: $t \in \{1, 2, ...\}$.

Money Demand:

$$\frac{M_t^d}{P_t} = \Phi(\overset{-}{i}_t, \overset{+}{Y}_t)$$

where

$$1 + i_t = (1 + r_t)\frac{\mathbb{E}_t P_{t+1}}{P_t}$$

For simplicity, assume:

$$\begin{aligned} Y_t &= Y \quad \forall t \geq 0 \\ r_t &= r \quad \forall t \geq 0 \end{aligned}$$

Then:

$$\frac{M_t^d}{P_t} = \Psi\left(\frac{\mathbb{E}_t \overset{-}{P}_{t+1}}{P_t}\right)$$

where $\Psi\left(\frac{\mathbb{E}_t P_{t+1}}{P_t}\right) \equiv \Phi\left((1 + r)\frac{\mathbb{E}_t P_{t+1}}{P_t} - 1, Y\right)$.

For simplicity, we assume that the function Ψ is linear:

$$\frac{M_t^d}{P_t} = c - b\frac{\mathbb{E}_t P_{t+1}}{P_t}$$

with $c > 0$ and $b > 0$.

Equilibrium requires:

$$M_t^d = M_t$$

where M_t is the money supply (in nominal terms).

Then:

$$\frac{M_t}{P_t} = c - b\frac{\mathbb{E}_t P_{t+1}}{P_t}$$

From the previous expression we get:

$$P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$$

where $\alpha \equiv \frac{b}{c} > 0$.

The expression above is a first-order stochastic linear difference equation in the price level.

Assume $c > b$, which implies

$$\alpha < 1$$

Then, we can solve $P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$ forward:

$$\begin{aligned}
P_t &= \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1} \\
&= \frac{M_t}{c} + \alpha \mathbb{E}_t \{ \frac{M_{t+1}}{c} + \alpha \mathbb{E}_{t+1} P_{t+2} \} \\
&= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \mathbb{E}_{t+1} P_{t+2} \\
&= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t P_{t+2} \quad \text{by LIE} \\
&= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \{ \frac{M_{t+2}}{c} + \alpha \mathbb{E}_{t+2} P_{t+3} \} \\
&= \frac{1}{c} M_t + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \frac{\alpha^2}{c} \mathbb{E}_t M_{t+2} + \alpha^3 \mathbb{E}_t \mathbb{E}_{t+2} P_{t+3} \\
&= \frac{1}{c} (M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2}) + \alpha^3 \mathbb{E}_t P_{t+3}
\end{aligned}$$

If we keep iterating forward we get:

$$P_t = \frac{1}{c}(M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2} + ... + \alpha^n \mathbb{E}_t M_{t+n}) + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Then:

$$P_t = \frac{1}{c} \sum_{i=0}^{n} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Letting $n \to \infty$:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} + \lim_{n \to \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

assuming that $\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \equiv \lim_{n \to \infty} \sum_{i=0}^{n} \alpha^i \mathbb{E}_t M_{t+i}$ is finite.

Suppose the price level satisfies:

$$\lim_{n \to \infty} \alpha^n \mathbb{E}_t P_{t+n} = 0$$

Then, the previous expression for P_t becomes:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \tag{1}$$

which is the so-called fundamental solution (no bubbles).

The expression above shows that the price level at t depends on the money supply expected to prevail from t into the indefinite future.

According to previous equation, if government deficits are to influence the price level path, it can only be through their effect on the expected path of base money. However, the government deficit and path of base money are not rigidly linked in any immutable way. The reason is that the government can, at least to a point, borrow by issuing interest-bearing government debt, and so need not necessarily issue base money to cover its deficit. To see this more precisely, we need to write down the budget constraint of the consolidated government:

$$G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1}) + \frac{M_t - M_{t-1}}{P_t} \quad (2)$$

with $M_0 > 0$, $D_0 > 0$ and $r_0 = r > 0$ given.

Under the system formed by equations (1) and (2), the inflationary consequences of government deficits depend sensitively on the government's strategy for servicing the debt that it issues.

Consider the following fiscal-monetary policy combination:

a) $G_t - T_t = \begin{cases} \overline{\Delta} & \text{if} \quad t \in \{1, 2, ..., T\} \\ \underline{\Delta} & \text{if} \quad t \in \{T+1, T+2, ...\} \end{cases}$

b) For $t \in \{1, 2, ..., T\}$, M_t grows at the constant exogenous rate θ and D_t evolves endogenoulsy to satisfy the budget constraint of the government;

c) For $t \in \{T+1, T+2, ...\}$, $D_t = D_T$ and M_t evolves endogenoulsy to satisfy the budget constraint of the government (if possible).

To find an equilibrium we work backwards in time.

We know that $r_{t-1} = r$, $G_t - T_t = \underline{\Delta}$ and $D_t = D_T$ $\forall t > T$.

Then, $G_t + r_{t-1} D_{t-1} = T_t + (D_t - D_{t-1}) + \frac{M_t - M_{t-1}}{P_t}$ becomes

$$\begin{aligned} G_t - T_t + r D_T &= (D_T - D_T) + \frac{M_t - M_{t-1}}{P_t} \quad \forall t > T \\ \underline{\Delta} + r D_T &= \frac{M_t - M_{t-1}}{P_t} \quad \forall t > T \qquad (3) \end{aligned}$$

We guess that there is an equilibrium with a constant growth of the money supply after T :

$$M_{t+1} = (1 + \mu) M_t \quad \forall t \geq T$$

where M_T is predetermined.

Then:

$$M_{t+i} = (1 + \mu)^i M_t \quad \forall t \geq T \qquad (4)$$

with $i \in \{0, 1, 2, ...\}$.

Substituting (4) into (1) we get:

$$
\begin{aligned}
P_t &= \frac{1}{c}\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t\{(1+\mu)^i M_t\} \quad \forall t \geq T \\
P_t &= \frac{M_t}{c}\sum_{i=0}^{\infty} [\alpha(1+\mu)]^i \\
P_t &= \frac{M_t}{c}\frac{1}{1-\alpha(1+\mu)}
\end{aligned}
$$

where we have assumed $\alpha(1+\mu) < 1$.

Then:

$$
P_t = \frac{1}{c\,[1-\alpha(1+\mu)]} M_t \quad \forall t \geq T \tag{5}
$$

Remark 1 $\frac{M_t}{P_t} = c\,[1-\alpha(1+\mu)]$ $\forall t \geq T$

Remark 2. $\pi_t = \mu \quad \forall t > T$

Now we consider $t \leq T$.

From $P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i}$ we get:

$$
\begin{aligned}
P_t &= \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \frac{1}{c} \sum_{i=T-t+1}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \quad \forall t \leq T \\
P_t &= \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \frac{1}{c} \alpha^{T+1-t} \sum_{i=T+1-t}^{\infty} \alpha^{i-(T+1-t)} \mathbb{E}_t M_{t+i} \\
P_t &= \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \frac{1}{c} \sum_{i=T+1-t}^{\infty} \alpha^{i-(T+1-t)} \mathbb{E}_t \{\mathbb{E}_{T+1} M_{t+i}\} \quad \text{LIE} \\
P_t &= \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \frac{1}{c} \sum_{j=0}^{\infty} \alpha^j \mathbb{E}_t \{\mathbb{E}_{T+1} M_{T+1+j}\} \\
P_t &= \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \mathbb{E}_t \left\{ \frac{1}{c} \sum_{j=0}^{\infty} \alpha^j \mathbb{E}_{T+1} M_{T+1+j} \right\}
\end{aligned}
$$

Using (1):

$$
P_t = \frac{1}{c} \sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{T+1-t} \mathbb{E}_t P_{T+1} \quad \forall t \leq T \qquad (6)
$$

We know $M_{t+1} = (1+\theta)M_t \ \forall t \in \{1, ..., T\}$, with $M_1 = (1+\theta)M_0 > 0$ given. Then:

$$M_{t+i} = (1+\theta)^i M_t \quad \forall t \in \{1, ..., T\} \tag{7}$$

where $i \in \{0, 1, ..., T-t\}$.

Substituting (5) and (7) into (6):

$$\begin{aligned}
P_t &= \frac{1}{c}\sum_{i=0}^{T-t} \alpha^i \mathbb{E}_t\left\{(1+\theta)^i M_t\right\} + \alpha^{T+1-t}\mathbb{E}_t\left\{\frac{M_{T+1}}{c\left[1-\alpha(1+\mu)\right]}\right\} \\
P_t &= \frac{M_t}{c}\sum_{i=0}^{T-t}\left[\alpha(1+\theta)\right]^i + \frac{\alpha^{T+1-t}M_{T+1}}{c\left[1-\alpha(1+\mu)\right]} \\
P_t &= \frac{M_t}{c}\sum_{i=0}^{T-t}\left[\alpha(1+\theta)\right]^i + \frac{\alpha^{T+1-t}(1+\mu)M_T}{c\left[1-\alpha(1+\mu)\right]} \\
P_t &= \frac{M_t}{c}\frac{1-\left[\alpha(1+\theta)\right]^{T+1-t}}{1-\alpha(1+\theta)} + \frac{\alpha^{T+1-t}(1+\mu)M_T}{c\left[1-\alpha(1+\mu)\right]} \quad \forall t \leq T
\end{aligned}$$

But $M_T = (1+\theta)^{T-t} M_t$. Then:

$$
\begin{aligned}
P_t &= \frac{M_t}{c}\frac{1-[\alpha(1+\theta)]^{T+1-t}}{1-\alpha(1+\theta)} + \frac{\alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t}M_t}{c\,[1-\alpha(1+\mu)]} \quad \forall t \leq T \\
P_t &= \left[\frac{1-[\alpha(1+\theta)]^{T+1-t}}{1-\alpha(1+\theta)} + \frac{\alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t}}{1-\alpha(1+\mu)}\right]\frac{M_t}{c} \\
P_t &= \frac{[1-\alpha(1+\mu)]\left[1-[\alpha(1+\theta)]^{T+1-t}\right] + [1-\alpha(1+\theta)]\left[\alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t}\right]}{[1-\alpha(1+\theta)]\,[1-\alpha(1+\mu)]}\frac{M_t}{c} \\
P_t &= \frac{1-[\alpha(1+\theta)]^{T+1-t} - \alpha(1+\mu) + \alpha(1+\mu)\,[\alpha(1+\theta)]^{T+1-t} + \alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t} - \alpha(1+\theta)\alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t}}{[1-\alpha(1+\theta)]\,[1-\alpha(1+\mu)]}\frac{M_t}{c} \\
P_t &= \frac{1-\alpha^{T+1-t}(1+\theta)^{T+1-t} - \alpha(1+\mu) + (1+\mu)\alpha^{T+2-t}(1+\theta)^{T+1-t} + \alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t} - (1+\mu)\alpha^{T+2-t}(1+\theta)^{T+1-t}}{[1-\alpha(1+\theta)]\,[1-\alpha(1+\mu)]}\frac{M_t}{c} \\
P_t &= \frac{1-\alpha^{T+1-t}(1+\theta)^{T+1-t} - \alpha(1+\mu) + \alpha^{T+1-t}(1+\mu)(1+\theta)^{T-t}}{[1-\alpha(1+\theta)]\,[1-\alpha(1+\mu)]}\frac{M_t}{c} \\
P_t &= \frac{1-\alpha(1+\mu) + [-(1+\theta)+(1+\mu)]\,\alpha^{T+1-t}(1+\theta)^{T-t}}{[1-\alpha(1+\theta)]\,[1-\alpha(1+\mu)]}\frac{M_t}{c}
\end{aligned}
$$

Then:

$$
P_t = \frac{1-\alpha(1+\mu) + (\mu-\theta)\,\alpha^{T+1-t}(1+\theta)^{T-t}}{[1-\alpha(1+\theta)]\,[1-\alpha(1+\mu)]}\frac{M_t}{c} \quad \forall t \leq T \tag{8}
$$

Remark: we still need to find the equilibrium value of μ.

Finding μ

Seigniorage:

$$\begin{aligned} S_t &\equiv \frac{M_t - M_{t-1}}{P_t} \\ &= \frac{M_t - M_{t-1}}{M_{t-1}} \frac{M_{t-1}}{P_t} \\ &= \frac{M_t - M_{t-1}}{M_{t-1}} \frac{M_{t-1}}{M_t} \frac{M_t}{P_t} \\ &= \frac{\mu_t}{1+\mu_t} \frac{M_t}{P_t} \end{aligned}$$

where $\mu_t \equiv \frac{M_t - M_{t-1}}{M_{t-1}}$ is the growth rate of the nominal money supply.

For $t > T$ we have $\mu_t = \mu$ and $\frac{M_t}{P_t} = c\,[1 - \alpha(1+\mu)]$. Then:

$$S_t = \frac{\mu}{1+\mu} c\,[1 - \alpha(1+\mu)] \qquad \forall t > T \qquad (9)$$

Substituting (9) into (3):

$$\underline{\Delta} + rD_T = \frac{\mu}{1+\mu} c\left[1 - \alpha(1+\mu)\right] \quad \forall t > T \quad (10)$$

To find μ from the equation above we first need to find the value of D_T, which will depend on μ (as we'll see below).

Finding D_T

From (2) we get: $G_t - T_t + r_{t-1}D_{t-1} = (D_t - D_{t-1}) + S_t \Rightarrow$

$$D_t = (1+r)D_{t-1} + \overline{\Delta} - S_t \quad \forall t \leq T$$

But we know $S_t \equiv \frac{M_t - M_{t-1}}{P_t} = \frac{\mu_t}{1+\mu_t}\frac{M_t}{P_t} \Rightarrow$

$$\begin{aligned} S_t &= \frac{\theta}{1+\theta}\frac{M_t}{P_t} \quad \forall t \leq T \\ &= \frac{\theta}{1+\theta}\frac{c\left[1-\alpha(1+\theta)\right]\left[1-\alpha(1+\mu)\right]}{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T+1-t}(1+\theta)^{T-t}} \quad \forall t \leq T \end{aligned}$$

where we used (8) in the second line.

Combining the previous two expressions we get:

$$D_t \quad = \quad (1+r)D_{t-1} + \overline{\Delta} - \frac{\theta}{1+\theta}\frac{c\left[1-\alpha(1+\theta)\right]\left[1-\alpha(1+\mu)\right]}{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T+1-t}(1+\theta)^{T-t}} \quad (11)$$

which holds $\forall t \leq T$.

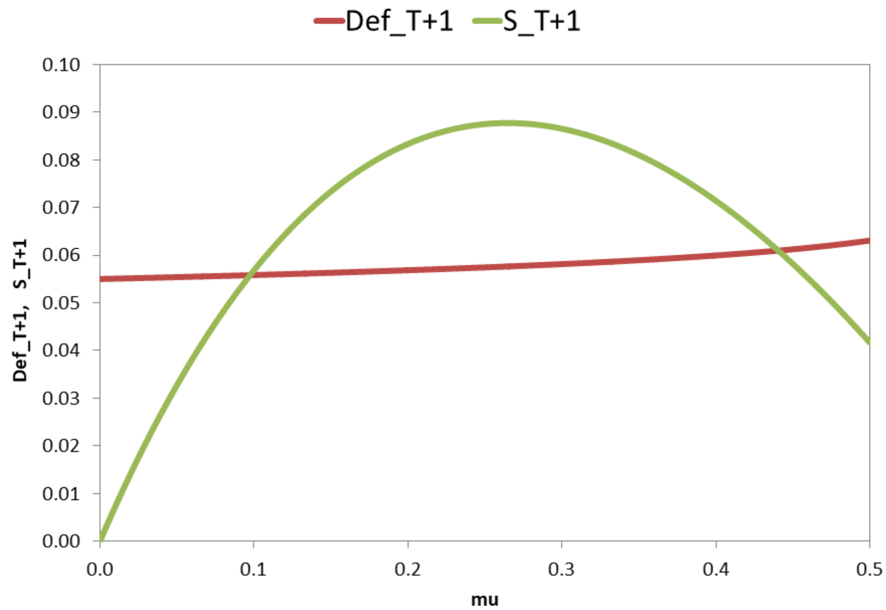
For any given μ, the expression above recursively determines the evolution of D_t from $t = 1$ until $t = T$, starting with D_0. For $t = T$ we get

$$D_T(\mu)$$

Substituting the expression above into (10) we get:

$$\underline{\Delta} + rD_T(\mu) = \frac{\mu}{1+\mu}c\left[1-\alpha(1+\mu)\right] \quad \forall t > T \qquad (12)$$

Graphically



Remark 1. Multple equilibria. We'll assume we are in the upward-sloping side of the seigniorage Laffer curve.

Remark 2. For a high-enough deficit, equilibrium will not exist.

Solution strategy:

(i) Choose a value for μ.

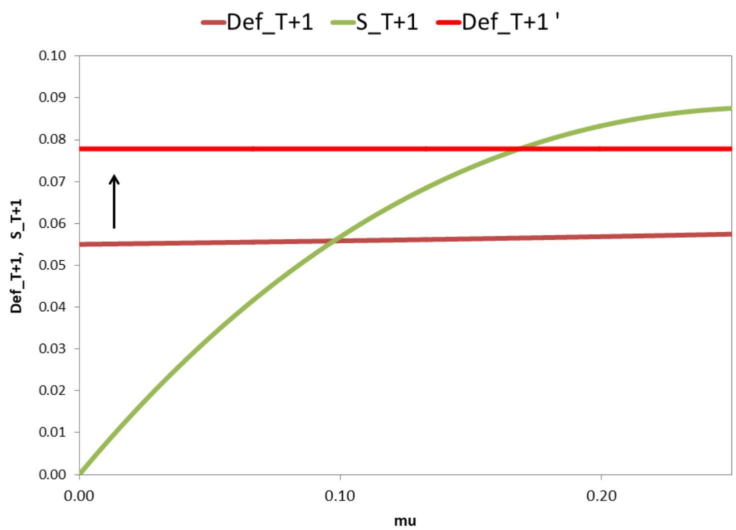
(ii) Iterate on (11) to find $D_T(\mu)$.

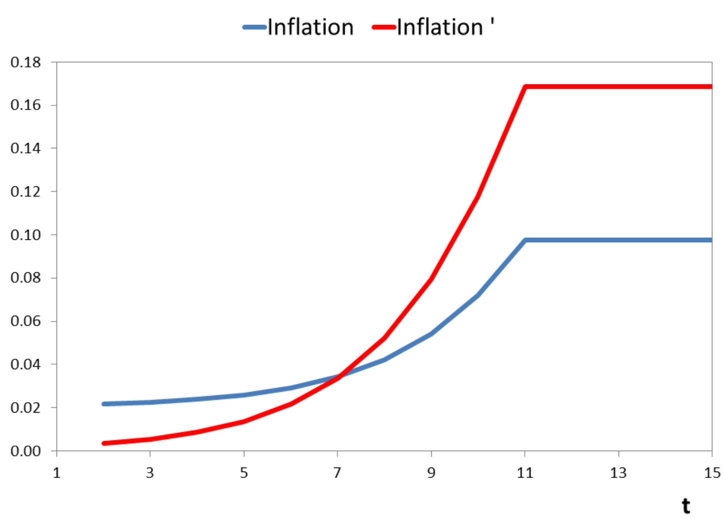
(iii) Compute $\underline{\mathbf{\Delta}} + rD_T(\mu) - \frac{\mu}{1+\mu}c\,[1 - \alpha(1+\mu)]$

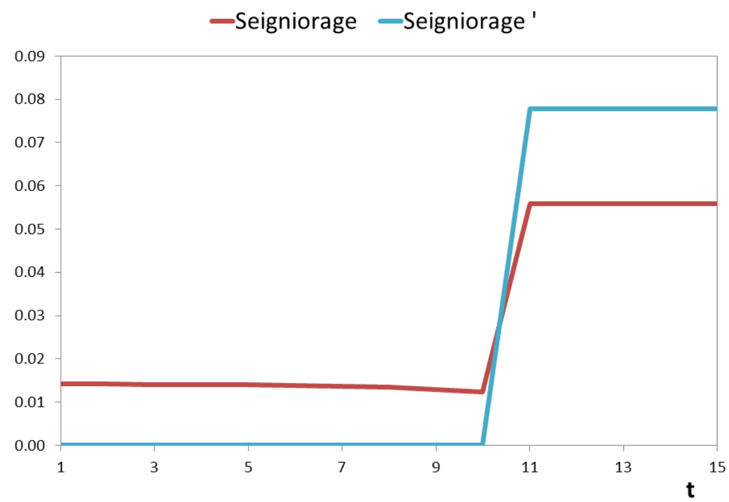
(iv) Change μ until the expression above equals 0, so (12) holds.

Unpleasant Monetarist Arithmetic

$$\downarrow \theta \Rightarrow \uparrow \mu$$



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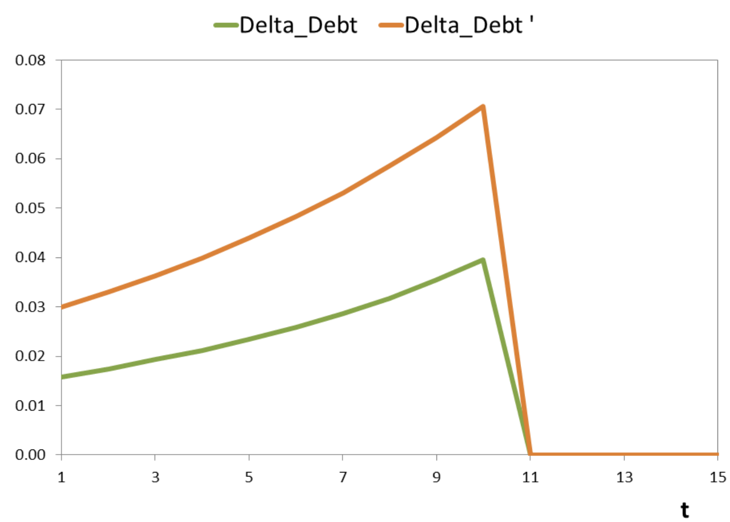
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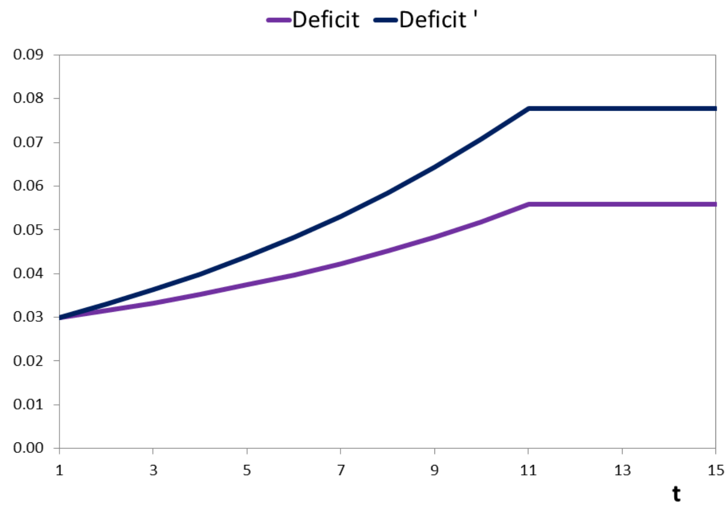
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Appendix I

From (6) we get:

$$\frac{P_t}{P_{t-1}} = \frac{\frac{1-\alpha(1+\mu)+(\mu-\theta)\alpha^{T+1-(t+1)}(1+\theta)^{T-(t+1)}}{[1-\alpha(1+\theta)][1-\alpha(1+\mu)]}\frac{M_{t+1}}{c}}{\frac{1-\alpha(1+\mu)+(\mu-\theta)\alpha^{T+1-t}(1+\theta)^{T-t}}{[1-\alpha(1+\theta)][1-\alpha(1+\mu)]}\frac{M_t}{c}} \quad \forall t \in \{1, ..., T-1\}$$

Then, $\forall t \in \{1, ..., T-1\}$:

$$\begin{aligned}
\frac{P_t}{P_{t-1}} &= \frac{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T+1-(t+1)}(1+\theta)^{T-(t+1)}}{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T+1-t}(1+\theta)^{T-t}}\frac{M_{t+1}}{M_t} \\
&= \frac{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T-t}(1+\theta)^{T-t-1}}{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T+1-t}(1+\theta)^{T-t}}(1+\theta) \\
&= \frac{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T-t}(1+\theta)^{T-t}}{1-\alpha(1+\mu)+(\mu-\theta)\,\alpha^{T+1-t}(1+\theta)^{T-t}}
\end{aligned}$$

Differentiating the expression above with respect to t we get:

$$\frac{\partial\left(\frac{P_t}{P_{t-1}}\right)}{\partial t}\begin{cases} >0 & \text{if} \quad \mu>\theta \\ =0 & \text{if} \quad \mu=\theta \\ <0 & \text{if} \quad \mu<\theta \end{cases}$$

Appendix II

From $G_t + rD_{t-1} = T_t + (D_t - D_{t-1}) + S_t$ we get:

$$(1+r)D_{t-1} = T_t - G_t + S_t + D_t$$

Then:

$$D_{t-1} = \frac{X_t}{1+r} + \frac{D_t}{1+r}$$

where $X_t \equiv T_t - G_t + S_t$.

Iterating forward:

$$\begin{aligned}
D_0 &= \frac{X_1}{1+r} + \frac{D_1}{1+r} \\
&= \frac{X_1}{1+r} + \frac{1}{1+r}\left(\frac{X_2}{1+r} + \frac{D_2}{1+r}\right) \\
&= \frac{X_1}{1+r} + \frac{X_2}{(1+r)^2} + \frac{D_2}{(1+r)^2} \\
&= \frac{X_1}{1+r} + \frac{X_2}{(1+r)^2} + \frac{1}{(1+r)^2}\left(\frac{X_3}{1+r} + \frac{D_3}{1+r}\right) \\
&= \frac{X_1}{1+r} + \frac{X_2}{(1+r)^2} + \frac{X_3}{(1+r)^3} + \frac{D_3}{(1+r)^3}
\end{aligned}$$

Then:

$$D_0 = \frac{X_1}{1+r} + \frac{X_2}{(1+r)^2} + ... + \frac{X_T}{(1+r)^T} + \frac{D_T}{(1+r)^T}$$

Then:

$$(1+r)D_0 = X_1 + \frac{X_2}{1+r} + ... + \frac{X_T}{(1+r)^{T-1}} + \frac{D_T}{(1+r)^{T-1}}$$

Then:

$$(1+r)D_0 = \sum_{t=1}^{T} \frac{X_t}{(1+r)^{t-1}} + \frac{D_T}{(1+r)^{T-1}}$$

Letting $T \to \infty$:

$$(1+r)D_0 = \sum_{t=1}^{\infty} \frac{X_t}{(1+r)^{t-1}} + \lim_{T\to\infty} \frac{D_T}{(1+r)^{T-1}}$$

Imposing $\lim_{T\to\infty} \frac{D_T}{(1+r)^{T-1}} = 0$ (TVC) and assuming $\sum_{t=1}^{\infty} \frac{X_t}{(1+r)^{t-1}}$ is finite:

$$(1+r)D_0 = \sum_{t=1}^{\infty} \frac{X_t}{(1+r)^{t-1}}$$

Then:

$$(1 + r)D_0 = \sum_{t=1}^{\infty} \frac{T_t - G_t + S_t}{(1 + r)^{t-1}}$$

Then:

$$\sum_{t=1}^{\infty} \frac{G_t - T_t}{(1 + r)^{t-1}} + (1 + r)D_0 = \sum_{t=1}^{\infty} \frac{S_t}{(1 + r)^{t-1}}$$

Notice that, for a given left-hand side, a reduction of S_t in some period requires an increase in S_t in some other period.

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